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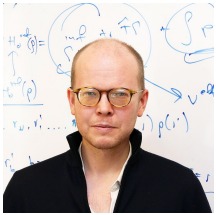
Kohn–Sham Inversion with Error Bounds

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Acknowledgements

In collaboration with M. Herbst[1,2] and A. Laestadius.[3,4]





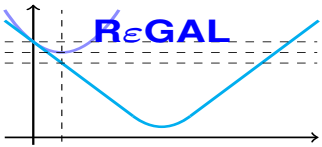
1 Mathematics for Materials Modelling, Institute of Mathematics & Institute of Materials, École Polytechnique Fédérale de Lausanne

2 National Centre for Computational Design and Discovery of Novel Materials (MARVEL), École Polytechnique Fédérale de Lausanne

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Outline

1 **Background**
- Density-Functional Theory
- Kohn–Sham Problem
- Moreau–Yosida Regularisation

2 **Inversion Scheme**
- Model Setting
- Exchange-Correlation Potential
- The Inversion Algorithm

3 **Results**
- Bulk Materials
- Error Bounds

4 **Conclusions**

5 **References**

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Density-Functional Theory

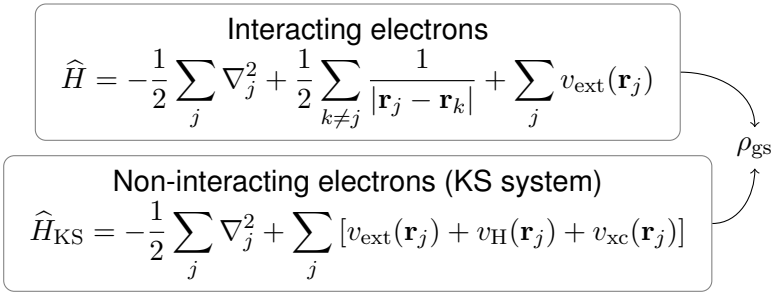
$$\widehat{H} = \underbrace{-\frac{1}{2}\sum_j \nabla_j^2 + \frac{1}{2}\sum_{k\neq j}\frac{1}{|\mathbf{r}_j - \mathbf{r}_k|}}_{\widehat{H}_0} + \sum_j v(\mathbf{r}_j)$$

$$\begin{aligned} E(v) &= \inf_{\psi} \langle\psi|\widehat{H}|\psi\rangle \\ &= \inf_{\rho} \{F(\rho) + \langle v,\, \rho\rangle\} \end{aligned}$$

$$\begin{aligned} F(\rho) &= \sup_{v} \{E(v) - \langle v,\, \rho\rangle\} \\ &= \inf_{\Gamma\mapsto\rho} \operatorname{Tr} \widehat{H}_0\Gamma \end{aligned}$$

$$\langle v,\, \rho\rangle = \int v\rho$$

The Kohn–Sham Approach

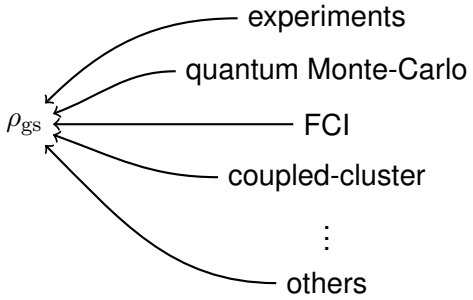


exchange correlation potential, $v_{\text{xc}} \longleftarrow$ unknown

The forward problem:
Given an approximate v_{xc}, determine ρ_{gs}

The Inverse Kohn–Sham Problem

The reverse problem:
Given a ρ_{gs}, determine v_{xc}



Better understanding of $F(\rho)$

Novel approximations to $F(\rho)$

Moreau–Yosida Regularisation

Densities: $\rho \in \mathcal{D}$ Potentials: $v \in \mathcal{V} = \mathcal{D}^*$

Definition

Let $\mathcal{D}$ be uniformly convex and $F : \mathcal{D} \to \mathbb{R} \cup \{+\infty\}$ convex and lower semicontinous functional. For some $\varepsilon > 0$, the *Moreau–Yosida regularisation* of F is

$$F^{\varepsilon}(\rho) = \inf_{\sigma \in \mathcal{D}} \left\{ F(\sigma) + \tfrac{1}{2\varepsilon} \|\sigma - \rho\|_{\mathcal{D}}^2 \right\}.$$

An infimal convolution

A Lossless Regularisation

$$E^{\varepsilon}(v) = \inf_{\rho \in \mathcal{D}} \{F^{\varepsilon}(\rho) + \langle v, \rho \rangle\}$$

$$E^{\varepsilon}(v) = E(v) - \frac{\varepsilon}{2}\|v\|_{\mathcal{V}}^2.$$

The regularisation retains the ground state energy

Consequence of infimal convolution and $E(\text{concave}) \leftrightarrow F(\text{convex})$.

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Inversion Scheme

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- Bulk Materials
- Error Bounds

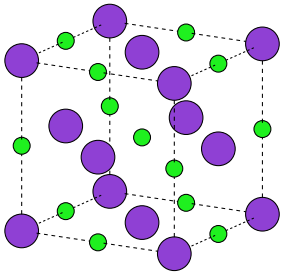
4 Conclusions

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Periodic Systems

$$\mathcal{D} = H^{-1}_{\text{per}}(\Omega, \mathbb{C}) \qquad \mathcal{V} = \mathcal{D}^* = H^1_{\text{per}}(\Omega, \mathbb{C})$$



Periodic Sobolev spaces [1]

$$\|u\|^2_{H^s_{\text{per}}} = \sum_{\mathbf{G}} (1 + |\mathbf{G}|^2)^s |\widehat{u}_{\mathbf{G}}|^2$$

The *duality mapping* $J : \mathcal{D} \to \mathcal{V}$,

$$J[\rho](\mathbf{r}) = (\Phi * \rho)(\mathbf{r}) = \int_{\mathbb{R}^3} \frac{\rho(\mathbf{x})}{4\pi|\mathbf{r} - \mathbf{x}|} e^{-|\mathbf{r}-\mathbf{x}|} \, \mathrm{d}\mathbf{x}$$

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Model Functional

Recall

$$\widehat{H}_{\mathrm{KS}} = -\frac{1}{2}\sum_j \nabla_j^2 + \sum_j \left[v_{\mathrm{ext}}(\mathbf{r}_j) + v_{\mathrm{H}}(\mathbf{r}_j) + v_{\mathrm{xc}}(\mathbf{r}_j)\right]$$

Fix $\rho_{\mathrm{gs}} \in \mathcal{D}$ and

$$\mathcal{F}(\rho) = T(\rho) + E_{\mathrm{H}}(\rho) + \int_\Omega v_{\mathrm{ext}}\rho$$

Minimise

$$\mathcal{E}(\rho\,;\rho_{\mathrm{gs}}) = \mathcal{F}(\rho) + \tfrac{1}{2\varepsilon}\|\rho - \rho_{\mathrm{gs}}\|_{\mathcal{D}}^2 \quad \text{over} \quad \rho \in \mathcal{D} \tag{1}$$

Proximal density:

$$\rho_{\mathrm{gs}}^{\varepsilon} = \operatorname*{argmin}_{\rho\in\mathcal{D}} \mathcal{E}(\rho, \rho_{\mathrm{gs}})$$

Unique minimiser of $\mathcal{E}$ [2]

The Exchange-Correlation Potential

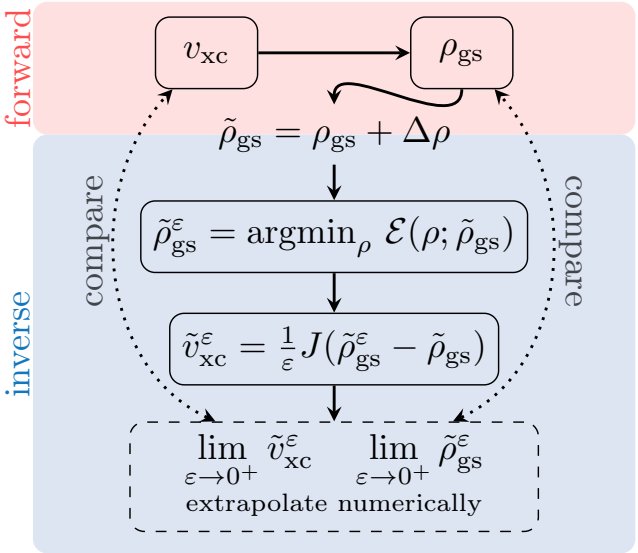
Since $\mathcal{F}^\varepsilon(\rho)$ is differentiable,

$$v_{\mathrm{xc}}^\varepsilon = -\frac{\delta \mathcal{F}^\varepsilon}{\delta \rho}[\rho_{\mathrm{gs}}^\varepsilon] = \frac{1}{\varepsilon} J[\rho_{\mathrm{gs}}^\varepsilon - \rho_{\mathrm{gs}}]$$

Exchange-correlation potential [3, 4]

$$\begin{aligned} v_{\mathrm{xc}}(\mathbf{r}) &= \lim_{\varepsilon \to 0^+} v_{\mathrm{xc}}^\varepsilon \\ &= \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} J\left[\rho_{\mathrm{gs}}^\varepsilon - \rho_{\mathrm{gs}}\right](\mathbf{r}) \\ &= \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \int_{\mathbb{R}^3} \frac{\rho_{\mathrm{gs}}^\varepsilon(\mathbf{x}) - \rho_{\mathrm{gs}}(\mathbf{x})}{4\pi|\mathbf{r}-\mathbf{x}|} e^{-|\mathbf{r}-\mathbf{x}|} \, \mathrm{d}\mathbf{x} \end{aligned}$$

The Inversion Algorithm



Results

1 Background
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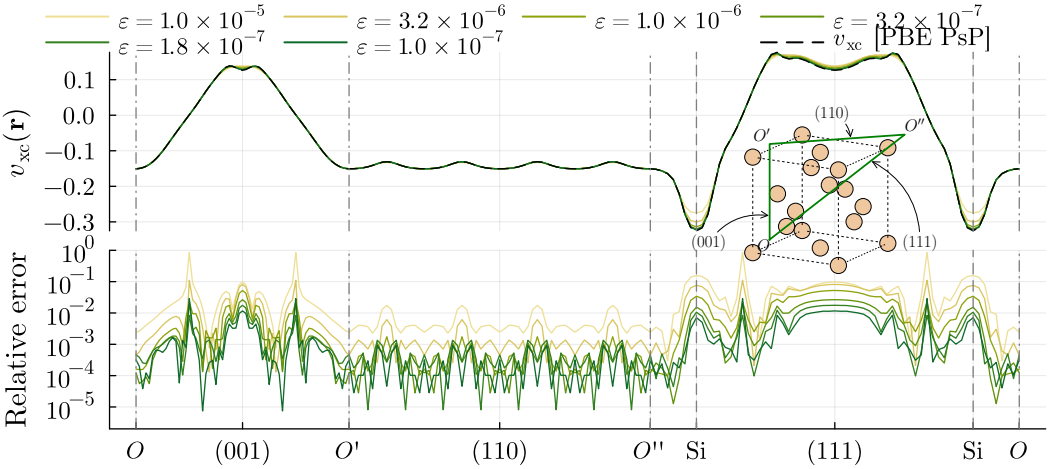
3 Results
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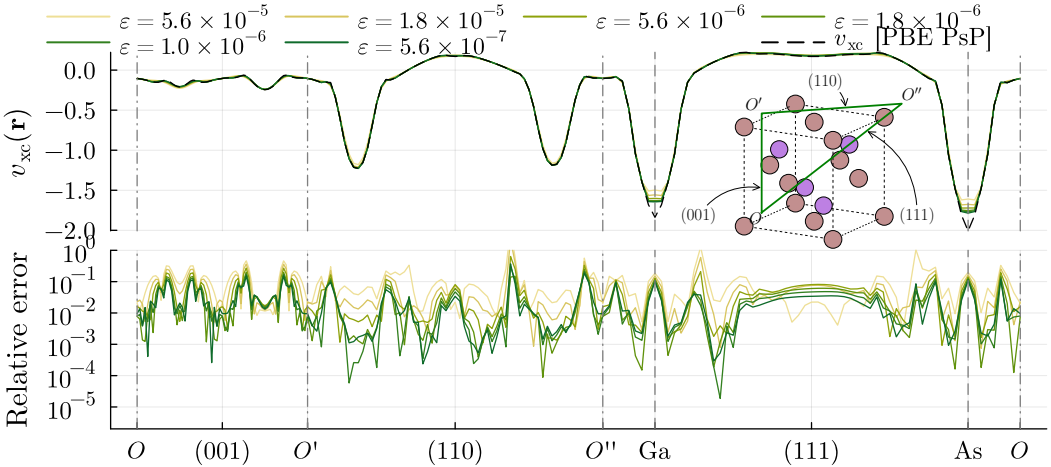
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Bulk Silicon

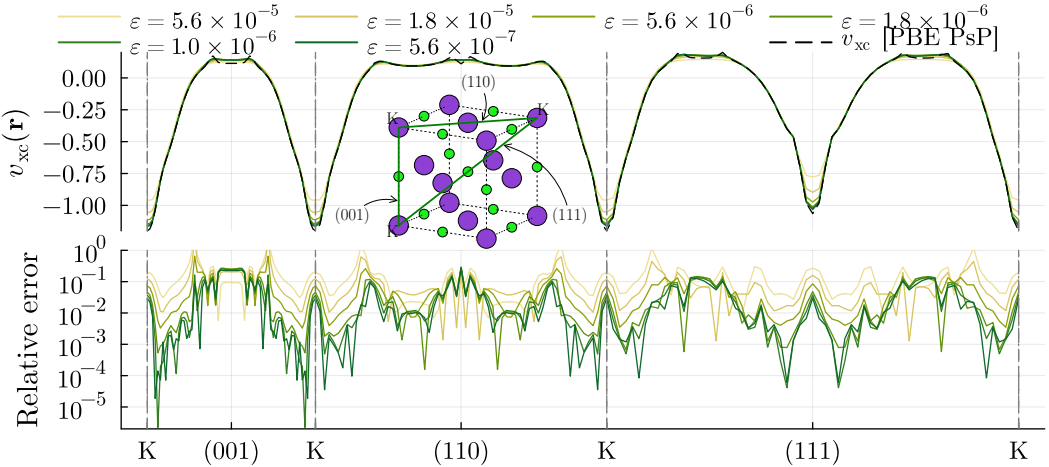


Implementation on github.com/mfherbst/supporting-my-inversion

Gallium Arsenide



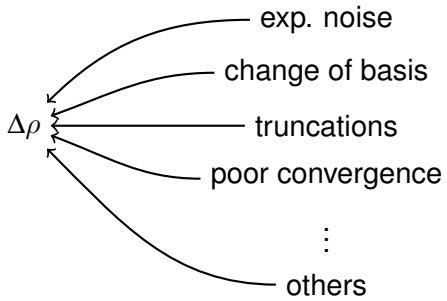
Potassium Chloride



Error Bounds

Inexact density:

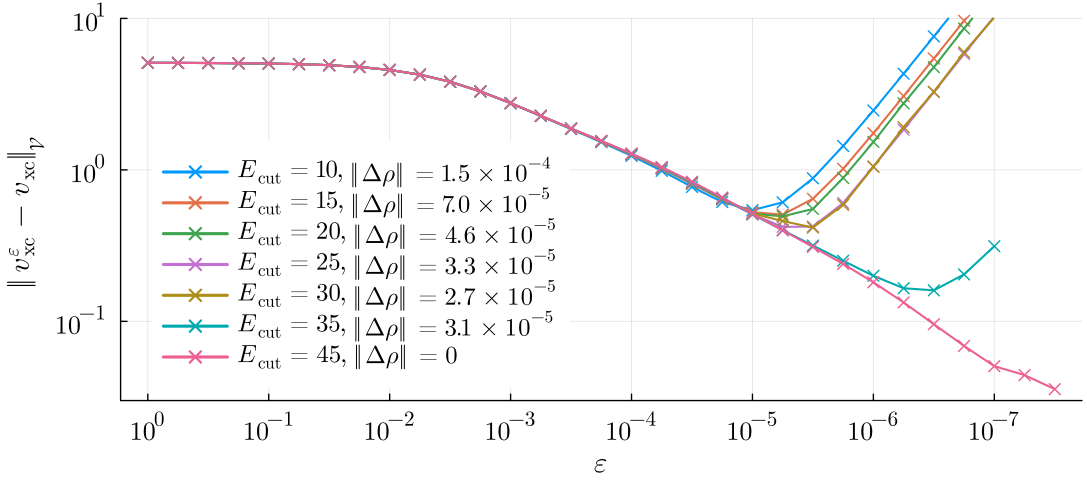
$$\widetilde{\rho}_{\mathrm{gs}} = \rho_{\mathrm{gs}} + \Delta\rho$$



The *total error*

$$\|v_{\mathrm{xc}} - \widetilde{v}^{\varepsilon}_{\mathrm{xc}}\|_{\mathcal{V}} \leq \underbrace{\|v_{\mathrm{xc}} - v^{\varepsilon}_{\mathrm{xc}}\|_{\mathcal{V}}}_{\text{terminating at finite } \varepsilon} + \underbrace{\|v^{\varepsilon}_{\mathrm{xc}} - \widetilde{v}^{\varepsilon}_{\mathrm{xc}}\|_{\mathcal{V}}}_{\text{using inexact } \rho_{\mathrm{gs}}} \tag{2}$$

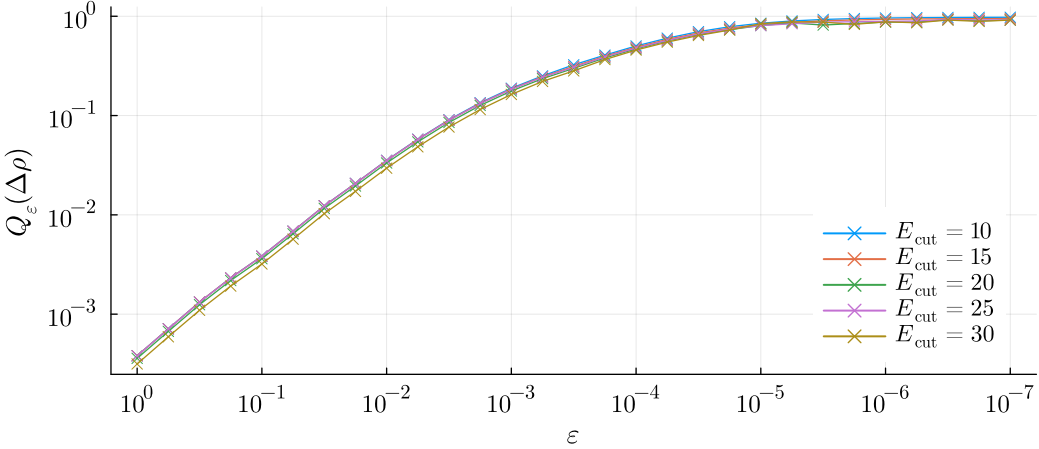
Convergence: Bulk Silicon



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Non-Expansiveness of $\rho \longmapsto \rho^{\varepsilon}$

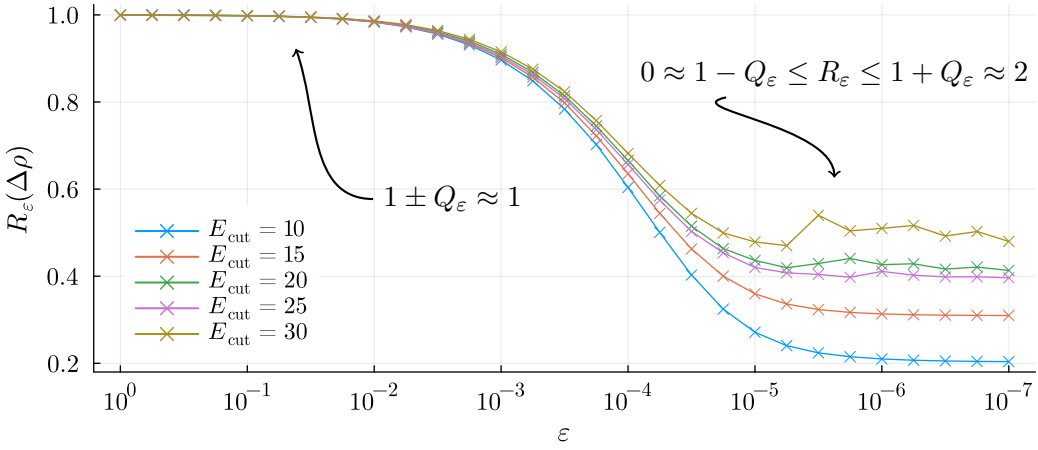
$$\left\|\rho^{\varepsilon}_{\mathrm{gs}} - \widetilde{\rho}^{\varepsilon}_{\mathrm{gs}}\right\|_{\mathcal{D}} \leq \|\rho_{\mathrm{gs}} - \widetilde{\rho}_{\mathrm{gs}}\|_{\mathcal{D}} = \|\Delta\rho\|_{\mathcal{D}} \qquad Q_{\varepsilon}(\Delta\rho) = \frac{\left\|\rho^{\varepsilon}_{\mathrm{gs}} - \widetilde{\rho}^{\varepsilon}_{\mathrm{gs}}\right\|_{\mathcal{D}}}{\|\Delta\rho\|_{\mathcal{D}}}$$



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Error Bounds: Inexact Densities

$$\|\widetilde{v}_{\mathrm{xc}}^{\varepsilon} - v_{\mathrm{xc}}^{\varepsilon}\|_{\mathcal{V}} \leq \frac{1 + Q_{\varepsilon}(\Delta\rho)}{\varepsilon}\|\Delta\rho\|_{\mathcal{D}} \qquad R_{\varepsilon}(\Delta\rho) = \varepsilon\frac{\|\widetilde{v}_{\mathrm{xc}}^{\varepsilon} - v_{\mathrm{xc}}^{\varepsilon}\|_{\mathcal{V}}}{\|\Delta\rho\|_{\mathcal{D}}}$$



Conclusions

- Mathematically rigorous inversion scheme
- Rigorous error estimates
- Practical use case of Moreau–Yosida regularisation
- First systems studied: Si, GaAs, KCl
- Implemented using `DFTK.jl` [5]



Outlook

- More complicated systems
- Reference densities from other sources
- Further error analysis
- Investigate proximal point algorithms

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References I

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2. **Barbu, V. & Precupanu, T.** *Convexity and Optimization in Banach Spaces.* 4th ed. ISBN: 9789400722477 (Springer Netherlands, 2012).
3. **Penz, M., Csirik, M. A. & Laestadius, A.** Density-potential inversion from Moreau–Yosida regularization. *Electron. Struct.* **5,** 014009 (2023).
4. **Herbst, M. F., Bakkestuen, V. H. & Laestadius, A.** *Kohn–Sham Inversion with Mathematical Guarantees.* 2025. arXiv: 2409.04372.
5. **.** github.com/mfherbst/supporting-my-inversion.

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Thank you for your attention!

Available on arXiv:2409.04372

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Motivation: The Differentiability of F

- The exact universal functional F is non-differentiable with respect to ρ [1].
- Practical implementations often assume differentiability, e.g., $v_{\mathrm{xc}} = \delta E_{\mathrm{xc}}/\delta\rho$.
- Regularisation of F to ensure differentiability.

[1] Lammert, P. E. Differentiability of Lieb functional in electronic density functional theory. Int. J. Quantum Chem. 107, 1943–1953 (10 2007)

Guiding Functional: Theory

Kinetic-only functional

$$T(\rho) = \sup_{v \in \mathcal{V}} \left\{ E^0(v) - \langle v, \rho \rangle \right\}$$

Periodic Hartree contribution

$$E_{\mathrm{H}}(\rho) = \sum_{\mathbf{G} \neq 0} \frac{|\widehat{\rho}_{\mathbf{G}}|^2}{|\mathbf{G}|^2}$$

Periodic Coulomb potential

$$v(\mathbf{r}) = \frac{4\pi}{|\Omega|} \sum_{\mathbf{G} \neq 0} \frac{1}{|\mathbf{G}|^2} e^{i\mathbf{G}\cdot\mathbf{r}}$$

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Analytic Error Bounds

Theoretically, $\forall\, \Delta\rho \in \mathcal{D}$ the ratios satisfies

$$0 \leq Q_\varepsilon(\Delta\rho) \leq 1$$

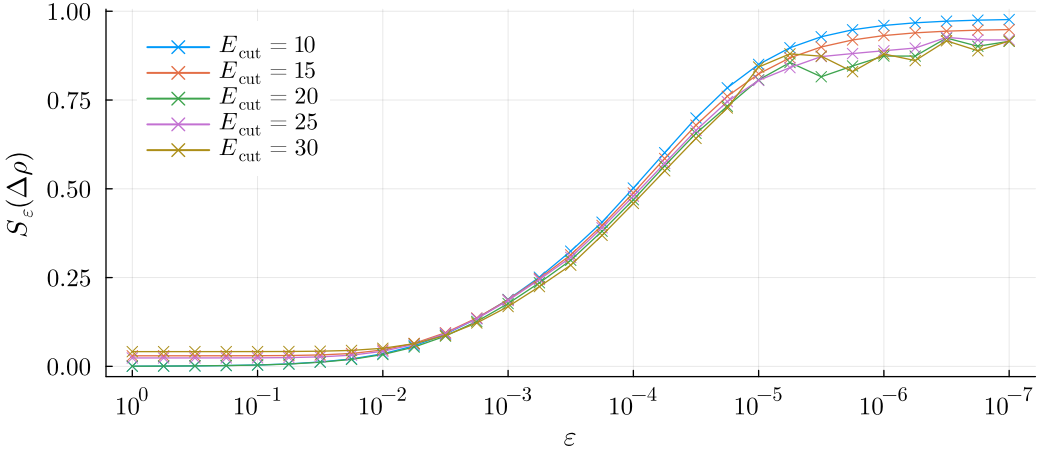
$$0 \leq 1 - Q_\varepsilon(\Delta\rho) \leq R_\varepsilon(\Delta\rho) \leq 1 + Q_\varepsilon(\Delta\rho) \leq 2$$

$$0 \leq S_\varepsilon(\Delta\rho) \leq Q_\varepsilon(\Delta\rho) \leq 1$$

$$\lim_{\varepsilon \to 0^+} Q_\varepsilon(\Delta\rho) = 1 \quad \text{and} \quad \lim_{\varepsilon \to 0^+} S_\varepsilon(\Delta\rho) \leq 1$$

Error Bounds: Inexact Densities

$$\left\|v_{\mathrm{xc}}^{\varepsilon}-\widetilde{v}_{\mathrm{xc}}^{\varepsilon}-\tfrac{1}{\varepsilon}J\left(\Delta\rho\right)\right\|_{\mathcal{V}}\leq\frac{1}{\varepsilon}Q_{\varepsilon}(\Delta\rho)\|\Delta\rho\|_{\mathcal{D}}\qquad S_{\varepsilon}(\Delta\rho)=\varepsilon\frac{\left\|v_{\mathrm{xc}}^{\varepsilon}-\widetilde{v}_{\mathrm{xc}}^{\varepsilon}-\tfrac{1}{\varepsilon}J\left(\Delta\rho\right)\right\|_{\mathcal{V}}}{\|\Delta\rho\|_{\mathcal{D}}}$$



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